

Is HP Turing Recognizable?

Reminder, $HP = \{ \langle M, w \rangle \mid M \text{ is valid TM encoding} \wedge M \text{ halts on } w. \}$

Turing Recognizable (TR)

A language L is TR if there exists a TM M such that for all strings $w \in L$, TM M halts & accepts. (May loop or reject other strings)

We show $HP \in TR$ by building the following recognizer using the UTM M_u .

$M_u(\langle M, w \rangle)$:
for $i = 1$ to ∞
 simulate M on input w for i steps
 if M halts then accept

Since we know HP is undecidable & also in TR, hence $HP \notin \text{co-TR}$.
(Otherwise it would be decidable).

Def

A TM E enumerates a language L if when E is run forever, E outputs a list of strings $x_1, x_2, \dots$ on the tape containing all & only the strings in L . (Repetition allowed). E is called an enumerator.

Claim A language is TR if & only if some enumerator enumerates it.

Proof

1. Assume enumerator E exists for L . Prove there exists a TM M that recognizes L .

Recognizer M for L .
$$\left\{ \begin{array}{l} M(w): \\ \quad \text{Run } E \\ \quad \text{for } i = 1 \text{ to } \infty \\ \quad \quad E \text{ prints string } s_i \\ \quad \quad \text{if } s_i = w \text{ halt \& accept} \end{array} \right.$$

2. Assume TM M recognizes L . Prove there exists an enumerator E for L .

Let $s_1, s_2, \dots$ be the list of strings in Σ^* .

E :
for $i = 1$ to ∞
 Run M for i steps on inputs $S = \{s_1, s_2, \dots, s_i\}$
 if any simulation of a string s_j halts & accepts output s_j

Both restrictions are crucial for a valid E .
 $j \in \{1, 2, \dots, i\}$

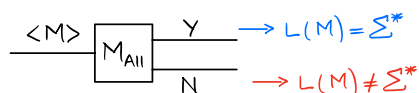
We have to avoid both, an infinite loop on a particular string & also a single simulation step on infinitely many strings.

REDUCTION

Let us see one in action first. Given the language

$All = \{ \langle M \rangle \mid M \text{ accepts all strings} \}$. We want to show All is undecidable.

Assume All is decidable. So, there exists M_{All} that decides it.

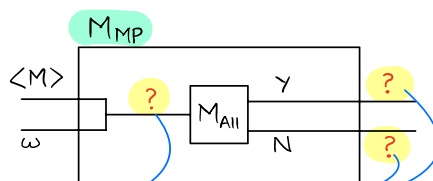


Now, we have to choose a language that is already known to be undecidable. Let us pick the Membership problem. (MP)

Our goal now is to show MP can be decided using M_{All} .

This results in a contradiction.

So, we want to build M_{MP} along following lines



We still need to figure out what happens here.

Argue that just providing $\langle M \rangle$ as input to M_{All} doesn't work.

We would be done if we could a TM $\langle M_w \rangle$ with following property.

$$\begin{aligned} L(M_w) = \Sigma^* &\Rightarrow w \in L(M) \\ \wedge \\ L(M_w) \neq \Sigma^* &\Rightarrow w \notin L(M) \end{aligned} \left\{ \begin{array}{l} \text{We can then just use the Y/N} \\ \text{answer of } M_{All} \text{ as answer of} \\ M_{MP}. \end{array} \right.$$

Consider the language of the following pieces of code: ($\Sigma = \{0, 1, \dots, 9\}$)

```
iseven(x):
  if x mod 2 == 0
    return accept
  else
    return reject
```

language is even integers

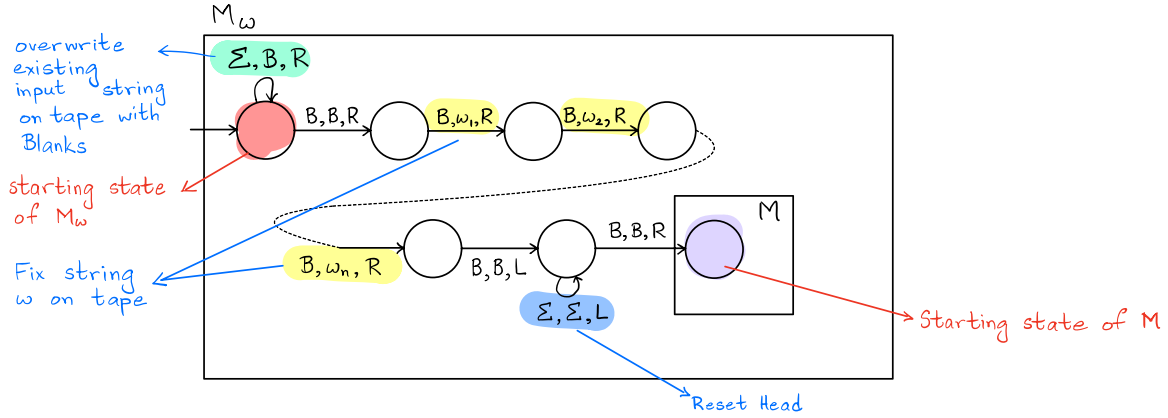
Modified
Code

```
iseven(x):
  x = 3
  if x mod 2 == 0
    return accept
  else
    return reject
```

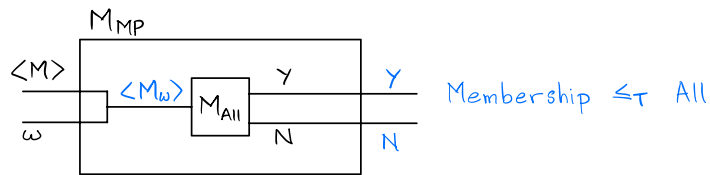
language is $\emptyset$
Setting $x = 2$ will make
language = Σ^*

We use similar idea for M_w .

$M_\omega(x)$:
 $x = \omega$
 Simulate M on ω



$$\begin{aligned} \omega \in L(M) &\Rightarrow L(M_\omega) = \Sigma^* \\ \omega \notin L(M) &\Rightarrow L(M_\omega) = \emptyset \end{aligned}$$



We have completed our reduction by building M_{MP} that decides Membership.

Membership $\leq_T$ All

Membership reduces to All

Pop Quiz

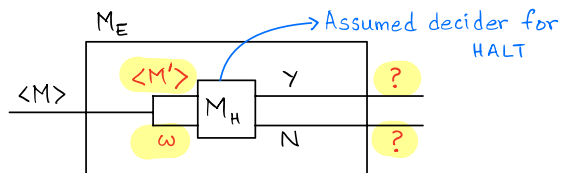
$$\text{All}_{\text{DFA}} = \{ \langle D \rangle \mid D \text{ is a valid DFA \& } L(D) = \Sigma^* \}$$

Show All_{DFA} is decidable.

Language A reduces to language B ($A \leq_T B$) if it is possible to decide A using an algorithm for B as an oracle / subroutine.

Let $\text{Empty}_{\text{TM}} = \{ \langle M \rangle \mid L(M) = \emptyset \}$.

Show $\text{Empty}_{\text{TM}} \leq_T \text{HALT}$



Given $\langle M \rangle$, we build $\langle M' \rangle$ as follows:

$\langle M' \rangle$:

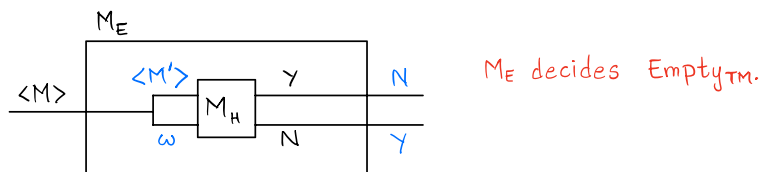
for $t=1$ to ∞

Simulate M on all strings $s \in \Sigma^*$ with $|s| \leq t$ for t steps
if any simulation halts & accepts then accept.

if $L(M) = \emptyset \Rightarrow M'$ does not halt

if $L(M) \neq \emptyset \Rightarrow M'$ halts

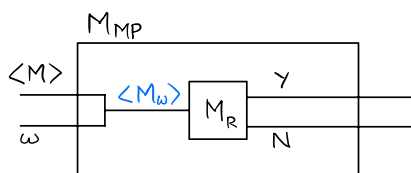
Note, M' does not look at its input. So can provide any w to M_H .



$\text{Reg}_{\text{TM}} = \{ \langle M \rangle \mid L(M) \text{ is regular} \}$.

Prove Reg_{TM} is undecidable.

We show $\text{Membership} \leq_T \text{Reg}_{\text{TM}}$

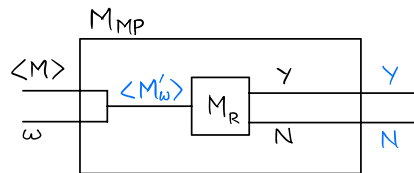


Check why M_w used for All_{TM} reduction doesn't work.

We build a modified M'_ω , that works as follows:

$M'_\omega(x)$:
 if x is of form $0^n 1^n$ accept
 else
 $x = \omega$
 Simulate M on ω

Your choice of a non-regular language



Now we have,

$\omega \in L(M)$	$\Rightarrow$	$L(M'_\omega) = \Sigma^*$	Regular
$\omega \notin L(M)$	$\Rightarrow$	$L(M'_\omega) = \{0^n 1^n\}$	Not regular